Trung Dao

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Research Interests

My research focuses on Vision-Language-Action (VLA) policies and small world action models for embodied agents: models that perceive their surroundings, reason about the task at hand, and act in real time on embodied hardware while remaining competitive with much larger systems. I am particularly interested in compact policy and world-model architectures, inference-efficient action generation, and quantization and deployment techniques that close the gap between large VLA models and the compute, latency, and power budgets of real robots. My background is in efficient generative modeling (diffusion distillation, one-step generation) and real-time, on-device perception and VLM deployment.

Education

University of Wisconsin-Madison — USA
Ph.D. in Computer Science (Advisor: Prof. Yong Jae Lee) — *Jan 2026 – Expected 2030*

Thang Long University — Vietnam
*Bachelor of Computer Science; GPA: 9.0/10.0 (**Valedictorian**)* — *Aug 2016 – Apr 2021*

Publications

(*) denotes equal contribution. Cited below as [P#].

[P1] Anh Nguyen, Ngan Nguyen, Duc Vu, **Trung Dao**, Viet Nguyen, Quan Dao, Kien Nguyen, Chi Tran, Phong Nguyen, Khoi Nguyen, Cuong Pham, Dimitris Metaxas, Vishal M. Patel, Anh Tran.
"Cross-Space Distillation: Teaching One-Step Students with Modern Diffusion Teachers." ECCV, 2026.

[P2] Anh Nguyen*, Viet Nguyen*, Duc Vu, **Trung Dao**, Chi Tran, Toan Tran, Anh Tran.
"Improved Training Technique for Shortcut Models." NeurIPS, 2025.

[P3] Viet Nguyen*, Viet Nguyen*, **Trung Dao**, Toan Tran, Anh Tran.
"SNOOPI: Supercharged One-step Diffusion Distillation with Proper Guidance." ICCV, 2025.

[P4] Quan Dao*, Hao Phung*, **Trung Dao**, Dimitris N. Metaxas, Anh Tran.
"Self-Corrected Flow Distillation for Consistent One-Step and Few-Step Image Generation." AAAI, 2025.

[P5] Hao Phung*, Quan Dao*, **Trung Dao**, Hoang Phan, Dimitris N. Metaxas, Anh Tran.
"DiMSUM: Diffusion Mamba – A Scalable and Unified Spatial-Frequency Method For Image Generation." NeurIPS, 2024.

[P6] **Trung Dao**, Thuan Nguyen, Thanh Le, Duc Vu, Khoi Nguyen, Cuong Pham, Anh Tran.
"SwiftBrushV2: Make Your One-step Diffusion Model Better Than Its Teacher." ECCV, 2024.

[P7] **Trung Dao***, Duc Hong Vu*, Cuong Pham, Anh Tran. "EFHQ: Multi-purpose ExtremePose-Face-HQ Dataset." CVPR, 2024.

Experience

University of Wisconsin-Madison — USA
Graduate Research and Teaching Assistant — *Jan 2026 – Present*

- **Research**: VLA policies and world action models for embodied agents, focusing on real-time, on-robot action generation that stays competitive with larger models.
- **Teaching**: Teaching Assistant for *Introduction to Computer Engineering* (Spring 2026).

Qualcomm AI Research — Vietnam
Staff Machine Learning Engineer — *Nov 2024 – Jan 2026*

- Fast-tracked promotion to **Staff Engineer** after a "Far Exceeds Expectations" (5/5) rating in the 2025 annual review.
- **On-Device Agentic AI**: Led the end-to-end quantization and deployment of InternVL3.5 (VLM) for always-on mobile screen understanding on Snapdragon — fine-tuned, quantized, and shipped a performant on-device inference flow under tight power/latency budgets with $< 5\%$ accuracy degradation.
- **Frontier Model Deployment**: Built the W4A16 quantization and deployment of LFM-2, the first hybrid transformer model to run on the Snapdragon Gen 5 chip, reaching **9000 tok/s** prefill and **90 tok/s** decode (4K context). Quantized and deployed further frontier models (Nemotron, InternVL) for resource-constrained edge devices.

- **Advanced Quantization**: Resolved critical W4A16 accuracy collapse in lightweight VLMs caused by activation/weight outliers, applying techniques (rotation, SpinQuant, AdaScale, GPTQ) selected per model and deployment to recover high performance on Snapdragon devices.
- **Efficient Diffusion Models**: Introduced a block-based distillation for FLUX.1 (40% smaller, $>$ 95% of HPSv2 retained); developed an attention-based text-encoder distillation enabling a T5-XXL $\rightarrow$ T5-base swap in PixArt-α with $<$ 2% HPSv2 drop.
- Joined via MovianAI as Senior Machine Learning Engineer; transitioned to Qualcomm after its acquisition (Apr 2025).

VinAI Research Vietnam
Research Resident *Mar 2023 – Oct 2024*

- **Advisor**: Dr. Anh Tran, Dr. Cuong Pham.
- **Research Focus**: Generative vision models, with an emphasis on GANs and diffusion models.
- **Selected works**:
  - Improved the quality of one-step and few-step text-to-image diffusion models [**P3, P4, P6**].
  - Introduced a novel diffusion architecture integrating Mamba for improved efficiency and scalability [**P5**].
  - Built a large-scale extreme-pose face dataset to improve synthesis quality and benchmark face recognition [**P7**].
- **HPC cluster management**: Managed and optimized a 48 $\times$ A100 GPU cluster, raising mean real-time GPU utilization from **15–20%** to **83%** via a novel preemptive queuing strategy.

VinAI Research Vietnam
AI Engineer *Dec 2020 – Mar 2023*

- **Advisor**: Dr. Dzung Nguyen, Dr. Anh Tran, Prof. Minh Hoai Nguyen.
- **Face Recognition Module** (Role: Module Owner)
  - Built multi-node training (up to 60M images) and a SLURM framework for profiling, tuning, and optimization; shipped models for masked access control and surveillance CCTV at **50K daily active identities**.
  - Achieved **8th** overall (**2nd** on Masked Dataset) at the ICCV21-MFR Competition, Jul 2022.
  - Quantized and deployed a 3-model module on Qualcomm AIC100 (up to 30 concurrent streams), plus NVIDIA (TensorRT) and Android (ONNX, MNN, NCNN); built supporting tools for visualization, video inference, and semi-automated data cleaning.
- **Face Detection Module** (Role: Module Co-owner)
  - Trained a multi-task masked-face detector for surveillance cameras, handling tiny faces, blocking artifacts, and occlusions; generated pseudo-masks via 2D/3D methods.
  - Co-built the AI SDK: optimized and deployed models on Xilinx with an asynchronous multi-stream (DeepStream) flow running up to **60 streams** on Xilinx ZCU104.
- **Traffic Sign/Light Recognition Module for Autonomous Driving** (Role: Module Co-owner)
  - Designed a CVAT-based pipeline to accelerate video labeling, yielding a dataset of *six superclasses and 317 child classes*; co-managed the labeling team for quality.
  - Built real-time perception for a moving autonomous-driving platform: a hierarchical multi-task recognizer (macro-F1 **98.3** on a private long-tailed 171-class set) with a ReID tracker for lighting robustness, deployed on-vehicle via TensorRT — the latency and robustness constraints embodied agents share.
- **Other**
  - **Noise Cancelling on Smartphone**: Converted models across various frameworks (PyTorch, TensorFlow, ONNX) into TFLite, followed by quantization and smartphone deployment. Optimized the existing algorithm with FFT, achieving a **40%** runtime reduction.

Honors and Awards

Summer Fellowship
University of Wisconsin-Madison *2026*

Top 5 Exceptional Vietnamese AI Talents
VNExpress AI Awards *2025*

Rank 2nd (Masked Dataset); 8th (Overall)
ICCV21-MFR Competition *2021*

Academic Excellence Scholarship
Thang Long University *2016 – 2021*

First Runner-up
VietAI Machine Learning Foundation Hanoi *2020*

First Runner-up
Fintech Track, Junction X Hanoi *2018*

Rank 76th
ICPC Asia Hanoi Regional Contest *2018*

Professional Services

Reviewer: CVPR (2023–2026), NeurIPS (2024, 2025), ECCV (2024, 2026), ICCV (2025), ICLR (2025), WACV (2025), ACCV (2022, 2024).

Outstanding Reviewer: CVPR 2026 (Top 5%).

Skills Summary

Expertise: Diffusion distillation and one-step generation, LLM/VLM quantization (W4A16, GPTQ, SpinQuant) and on-device deployment, real-time perception, multi-node training

Languages: Python, C++, Bash, SQL

Tools: PyTorch, TensorFlow, TFLite, TensorRT, AIMET, ONNX, NCNN, MNN, DeepStream, OpenCV, SLURM, Docker, Git

References

Prof. Yong Jae Lee: Susan Beth Horwitz Professor, University of Wisconsin-Madison: yongjaelee@cs.wisc.edu

Dr. Dzung Nguyen: Director, Engineering, Qualcomm AI Research, Vietnam: dungnt@qti.qualcomm.com

Dr. Anh Tran: Principal Engineer, Qualcomm AI Research, Vietnam: anhtra@qti.qualcomm.com

Prof. Minh Hoai Nguyen: Deputy Director of AIML, University of Adelaide, Australia: mh.nguyen@adelaide.edu.au